

AlgoPathFinder

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Letter of Transmittal

24th May , 2025

The Lecturer

Department of Computer Science and Engineering

Northern University Bangladesh (NUB)

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Subject : Letter of Transmittal.

Dear Sir ,

We are very pleased to present to you our lab report titled ‘Music Player’ as required by NUB for this course titled “ Software Development 3 (CSE 3292)”. It was indeed a great opportunity for us to work on this project to actualize our theoretical knowledge into practice.

Now we are looking forward to your kind appraisal of our report.

Finally, we would like to thank you for giving us the opportunity to pursue our studies with your guidance.

Yours sincerely,

Shahriar Nazim – 340

Md. Sayem Khan – 339

Rahnuma Tabassum Orpita – 341

Mahdin Islam – 405

Md Mithun Ahmed - 388

Student's Declaration

We hereby declare that this lab report titled '**AlgoPathFinder – Algorithm Simulation Website**' is our original work. It has not been submitted previously or concurrently for any other academic purpose, either by us or by any other individual. We affirm that the project and its documentation are free from plagiarism and data falsification. All external materials, references, or resources used in the development and writing of this report have been properly cited and acknowledged.

Abstract

AlgoPathFinder is an interactive web-based platform developed to visually simulate fundamental computer science algorithms, including pathfinding and sorting techniques. The primary goal of the project is to enhance algorithmic understanding through dynamic visualizations that demonstrate how algorithms operate step-by-step in real time. Designed with students and programming enthusiasts in mind, AlgoPathFinder offers an intuitive interface where users can select algorithms, adjust parameters, and observe how changes affect performance and outcomes. The simulation includes popular pathfinding algorithms like Dijkstra's and A*, along with sorting algorithms such as Bubble Sort, Selection Sort, and Quick Sort. The project leverages web technologies like HTML, CSS (TailwindCSS), and JavaScript to create smooth animations and responsive interactions. AlgoPathFinder serves as both an educational tool and a practical demonstration of how frontend technologies can be used to visualize complex concepts in computer science.

Acknowledgment

We would like to express our sincere gratitude to all those who supported and guided us throughout the development of this project titled "**AlgoPathFinder – Algorithm Simulation Website.**"

First and foremost, we are thankful to our respected faculty members and project supervisor(s) for their invaluable guidance, constant encouragement, and constructive feedback at every stage of this work. Their insights played a crucial role in shaping the direction of this project.

We also extend our heartfelt thanks to our institution for providing the resources and a supportive learning environment that allowed us to explore, experiment, and implement our ideas effectively.

Special thanks to our peers, friends, and family members for their continuous motivation, support, and patience during the completion of this project.

Lastly, we acknowledge the online programming communities and open-source contributors whose shared knowledge and tools significantly helped us during the development of this algorithm visualization platform.

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Chapter 1: Project overview

1.1 Introduction

An audio music player project involves creating software that enables users to play, manage, and enjoy music files on various devices. Typically, such projects incorporate features like play, pause, stop, volume control and often integrate with libraries or databases to organize and retrieve music tracks efficiently. The project's scope can range from a simple standalone player to a more sophisticated application with user-friendly interfaces across different platforms like desktop.

1.2 Project overview

AlgoPathFinder is a web-based educational platform designed to visually demonstrate the working of various algorithms, focusing on pathfinding and sorting. The main objective of this project is to bridge the gap between theoretical algorithm learning and practical understanding by providing an intuitive, interactive and engaging visualization tool.

In traditional algorithm education, learners often struggle to grasp how an algorithm operates step by step. AlgoPathFinder addresses this challenge by offering real-time animations of how data is processed, traversed, and sorted. Users can interact with the simulation by choosing different algorithms, adjusting speed, grid size, or array length, and observing how this change affects the algorithm behaviour.

The current version of the platform includes simulations for:

- Pathfinding Algorithms such as:
 - o Dijkstra's Algorithm
 - o A Search Algorithm
 - o Breadth-First Search (BFS)
 - o Depth-First Search (DFS)
- Sorting Algorithms such as:
 - o Bubble Sort

- o Selection Sort

The frontend is built using HTML, TailwindCSS, and JavaScript, ensuring a responsive and modern user interface. The animations are created using native JavaScript, which ensures high performance and better control over rendering. Additional enhancements such as color-coding, step indicators, and real-time metrics further help users understand the algorithmic process. This platform not only helps learners visualize how algorithms work but also allows them to experiment and compare their performances under different conditions. Ultimately, AlgoPathFinder aims to make algorithm learning more accessible, interactive, and fun.

1.3 Objectives

The main objective of the AlgoPathFinder project is to develop an interactive web application that visually demonstrates the step-by-step execution of various algorithms, making it easier for students and learners to understand their inner workings.

The specific objectives are:

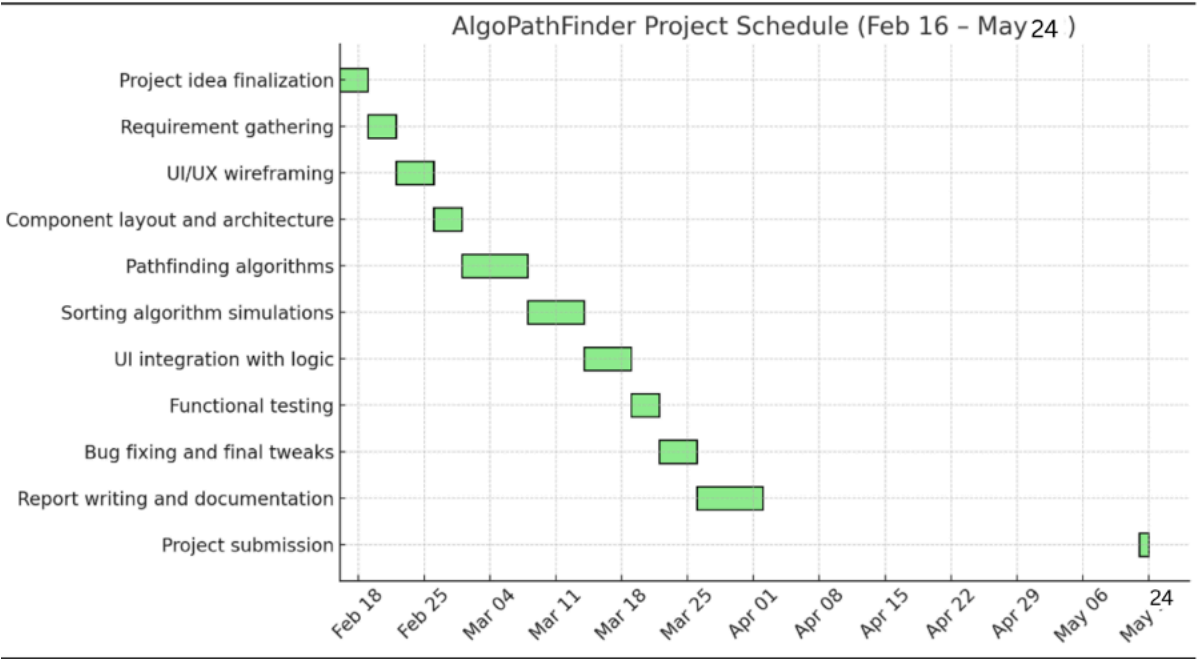
1. To visually simulate popular pathfinding and sorting algorithms in real-time.
2. To provide an interactive learning tool where users can experiment with different algorithms, input sizes, and configurations.
3. To enhance conceptual clarity by representing algorithm logic through animations and color-coded steps.
4. To create an intuitive and responsive user interface using modern web technologies.
5. To support algorithm comparison by helping users observe differences in efficiency and behavior under various conditions.
6. To promote self-learning and exploration in the field of data structures and algorithms.

1.4 Limitations Of The Project

While AlgoPathFinder provides a valuable tool for understanding algorithm behavior through visual simulation, there are a few limitations in its current implementation:

1. **Limited Algorithm Coverage:** The platform currently includes only a set of basic pathfinding and sorting algorithms. More complex algorithms like graph traversal, dynamic programming, or machine learning algorithms are not yet implemented.
2. **No Backend or User Authentication:** The current version is entirely frontend-based. Features like saving progress, user profiles, or history of simulations are not supported.
3. **Performance Constraints:** For very large data sets (e.g., big grids or long arrays), the visualizations may lag or slow down due to browser limitations and single-threaded JavaScript execution.
4. **Lack of Mobile Optimization:** While the interface is responsive to some extent, the visualization experience is best on larger screens. Some UI elements may not render optimally on mobile devices.
5. **No Step-by-Step Explanation:** The animations show how the algorithm works visually, but they do not provide textual explanations or pseudocode alongside the simulation (which could help with deeper understanding).
6. **Limited User Customization:** Users can control some parameters (like speed or size), but advanced customization options like defining custom heuristics or changing algorithm logic are not available.

Chapter 2: Project Scheduling Chart



Chapter 3: Project UI

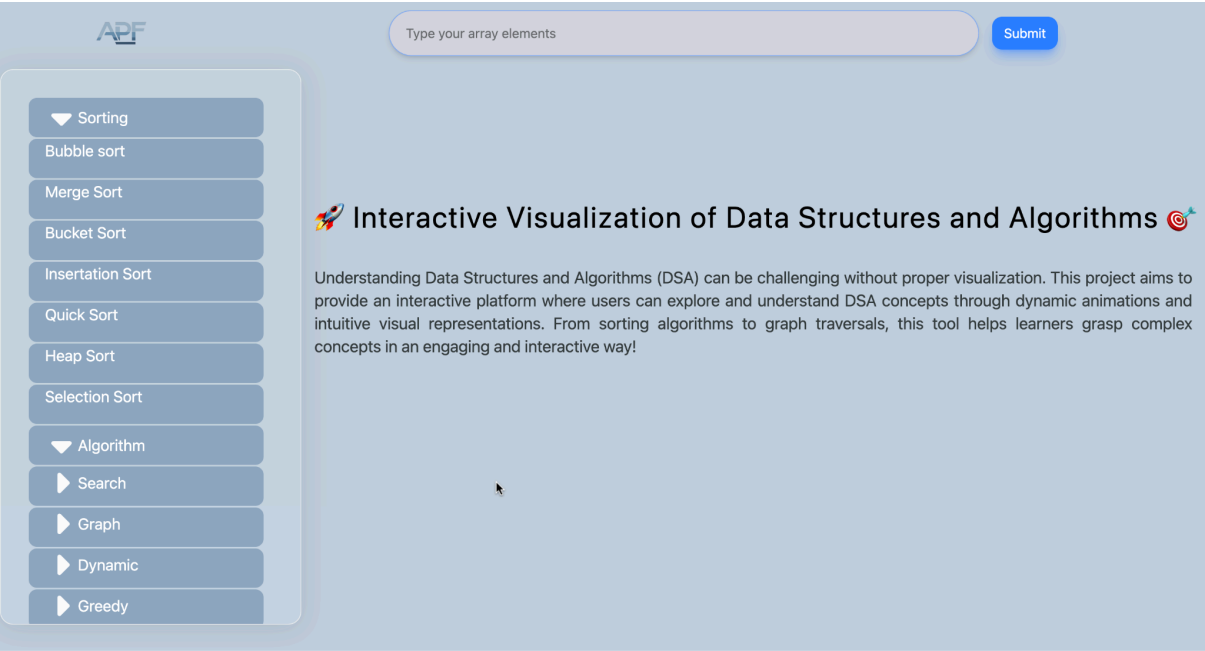


Figure 3.1 Starting page

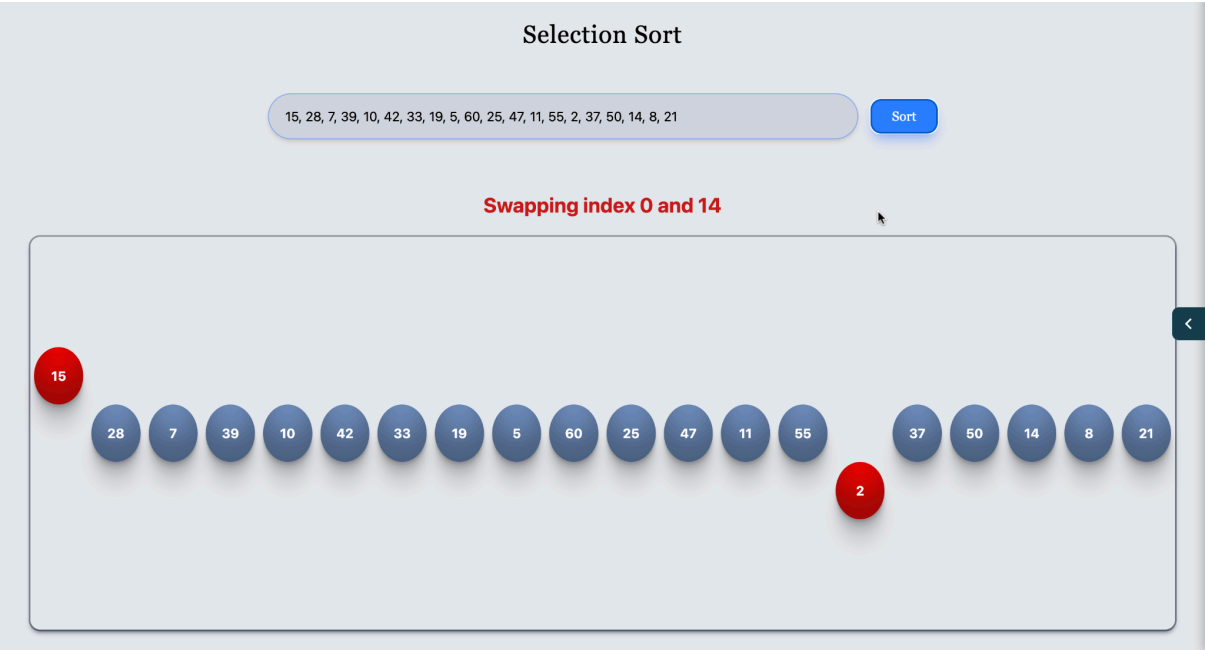


Figure 3.2 Selection Sort

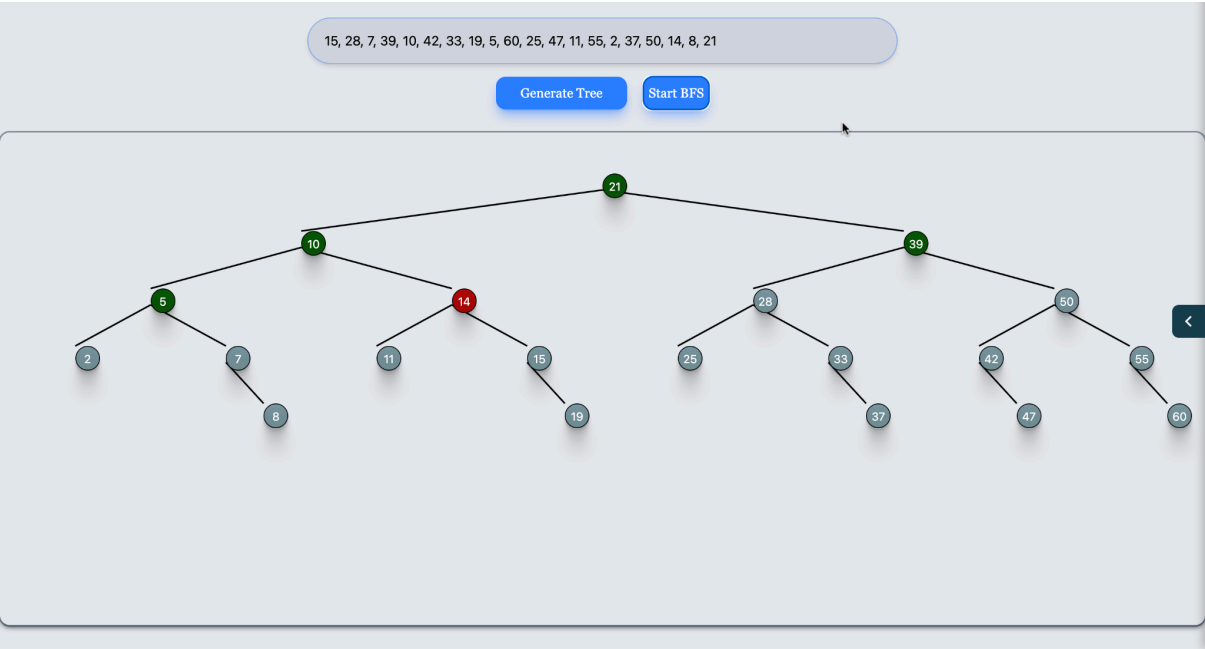


Figure 3.3 Breadth First Search

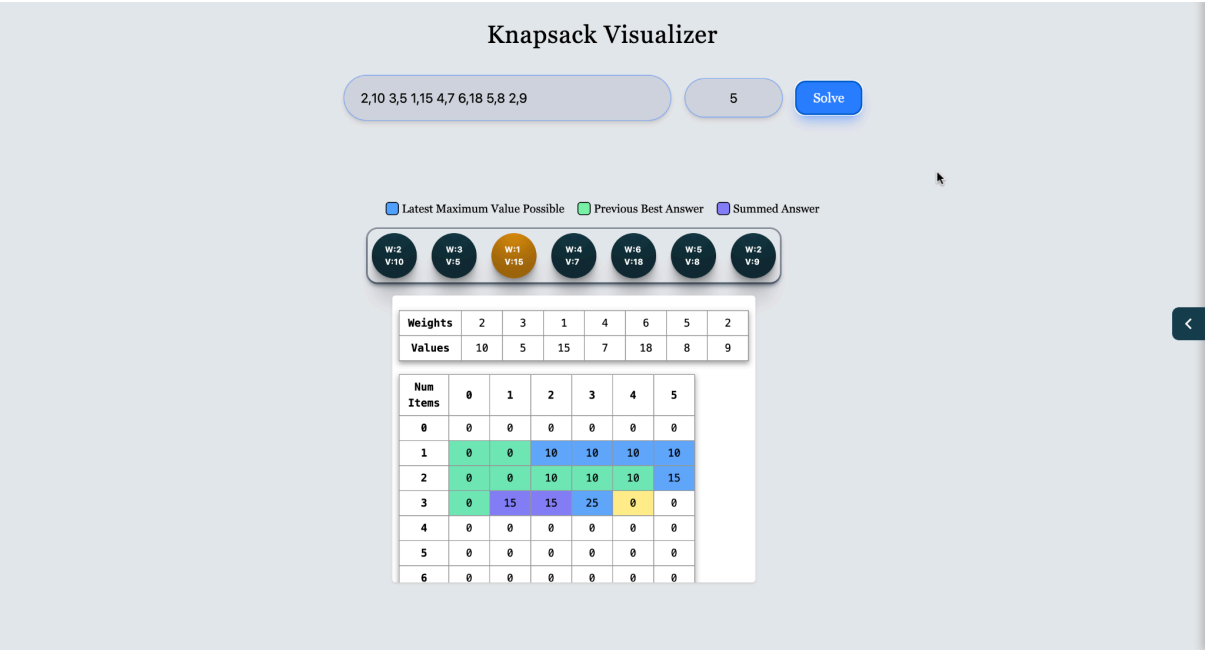


Figure 3.4 0/1 Knapsack

Chapter 4: Testing

4.1 Basic Functionality

1. Users can choose from a list of algorithms to simulate, including:

Pathfinding Algorithms: Dijkstra's, A*, BFS, DFS

Sorting Algorithms: Bubble Sort, Selection Sort, Quick Sort

2. Interactive Grid/Array

For pathfinding algorithms, users interact with a grid:

- Set start and end nodes
- Place walls/obstacles
- Visualize the traversal path and shortest route

For sorting algorithms, users see a dynamic bar chart that shows:

- How elements are compared
- Swapping steps
- Final sorted output

3. Control Panel

Users can:

- Adjust animation speed
- Reset the grid or array
- Pause/Resume the simulation
- Randomize data (for sorting)

4. Visualization

Real-time color-coded animations show algorithm progression:

- Visited nodes, frontier, and final path in pathfinding
- Comparisons and swaps in sorting
- Visual differences help in comparing algorithm efficiency

5. Multiple Runs

- Users can run different algorithms on the same setup to compare outcomes
- Quick reset options let users re-run simulations with different parameters

4.2 UI Responsiveness and Usability

- Built with TailwindCSS for responsive design across all screen sizes.
- Interactive UI with clickable grids, buttons, and sliders.
- Optimized for mobile, tablet, and desktop views.
- Color-coded animations provide visual feedback during simulations.
- Prevents errors with input validation (e.g., missing start/end points).
- Simple and clean layout ensures easy navigation for all users.
- Real-time controls for adjusting speed, resetting, or switching algorithms.
- Designed to be user-friendly for beginners learning algorithms.

4.3 Error Handling and Performance

Error Handling

- Prevents simulation if start or end nodes are missing.
- Disables or resets buttons when the app is in an invalid state.
- Provides visual cues (like color changes or disabled controls) for invalid actions.
- Optional alert messages guide users when input is incomplete or incorrect.

Performance

- Efficient algorithms ensure smooth simulations on moderate grid/array sizes.
- Uses optimized rendering to update only necessary UI elements.
- Performance may drop on very large grids or arrays due to browser and JavaScript limits.
- Lightweight design ensures fast loading and smooth usage even on mobile devices

Chapter 5: Conclusion

The AlgoPathFinder project successfully demonstrates how algorithm visualization can enhance understanding of complex concepts like pathfinding and sorting. By offering an interactive and responsive platform, it helps users — especially students — grasp how algorithms work step by step. The visual representation of each process, combined with user-friendly controls and real-time feedback, makes learning both engaging and effective. While there are areas for future improvement, such as expanding algorithm coverage and enhancing mobile optimization, the current version lays a strong foundation for educational simulation tools.

The AlgoPathFinder project successfully demonstrates how algorithm visualization can enhance understanding of complex concepts like pathfinding and sorting. By offering an interactive and responsive platform, it helps users — especially students — grasp how algorithms work step by step. The visual representation of each process, combined with user-friendly controls and real-time feedback, makes learning both engaging and effective. While there are areas for future improvement, such as expanding algorithm coverage and enhancing mobile optimization, the current version lays a strong foundation for educational simulation tools.

Reference

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